

COMPUTER NETWORKS

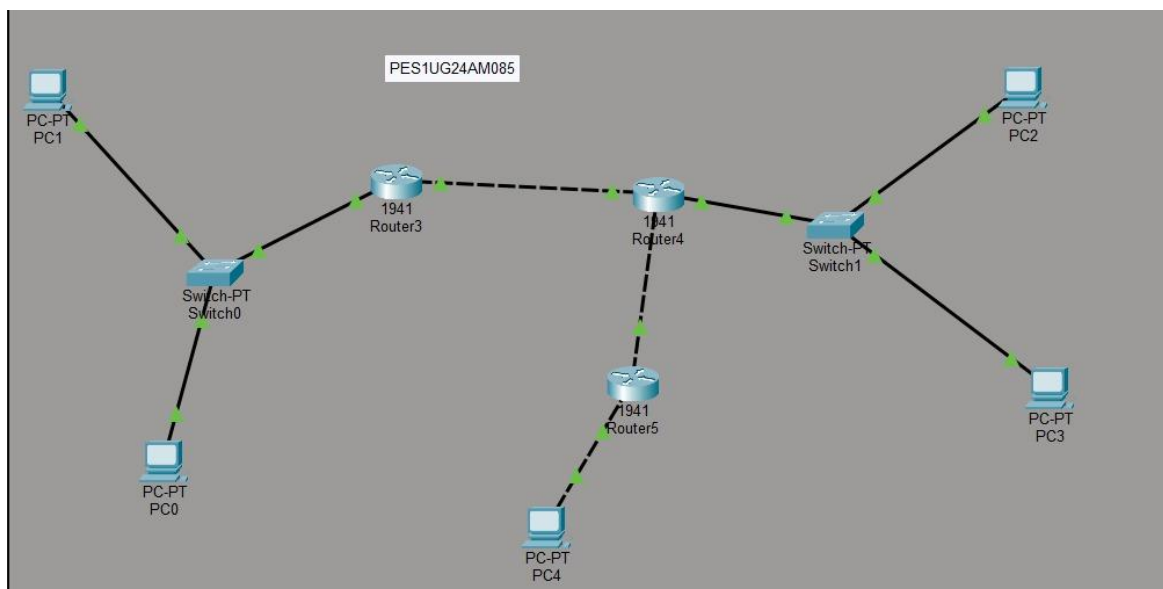
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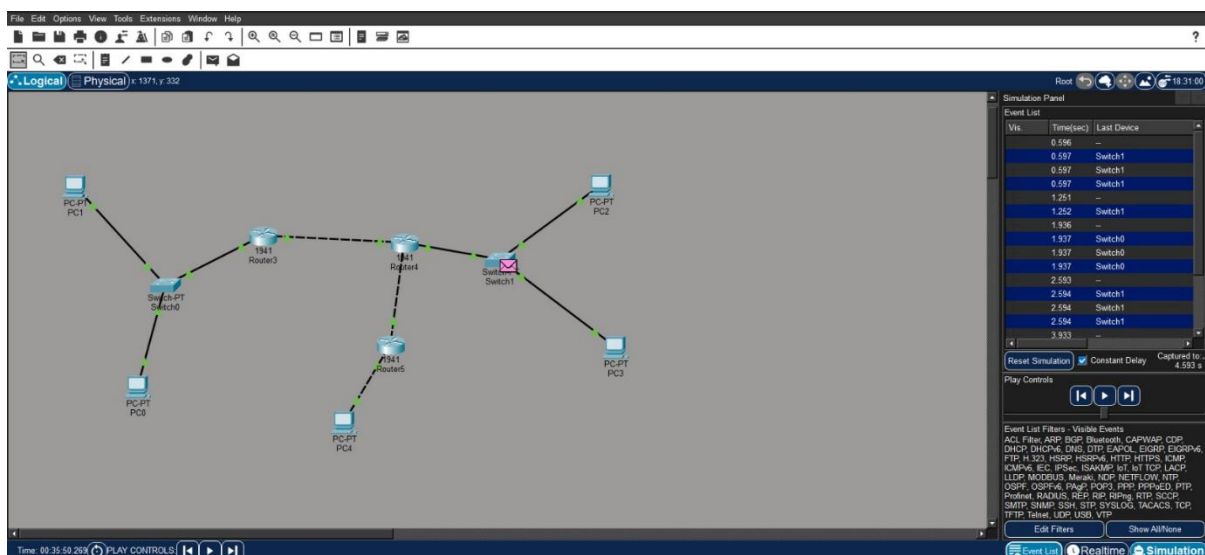
CLASS: 4B AIML

TASK (2A)

1.



2.



3. What is your observation and understanding of this concept that you have implemented?

From this experiment, I understood how multiple networks can be connected using routers and switches. PCs in the same LAN can communicate easily through a switch, but communication between different LANs is only possible using routers. By assigning IP addresses and configuring routing, all PCs were able to communicate with each other across different networks. This task helped me understand the real use of routers, interfaces, and routing in a network.

4. Why is it necessary to assign an IP address to the newly added interface?

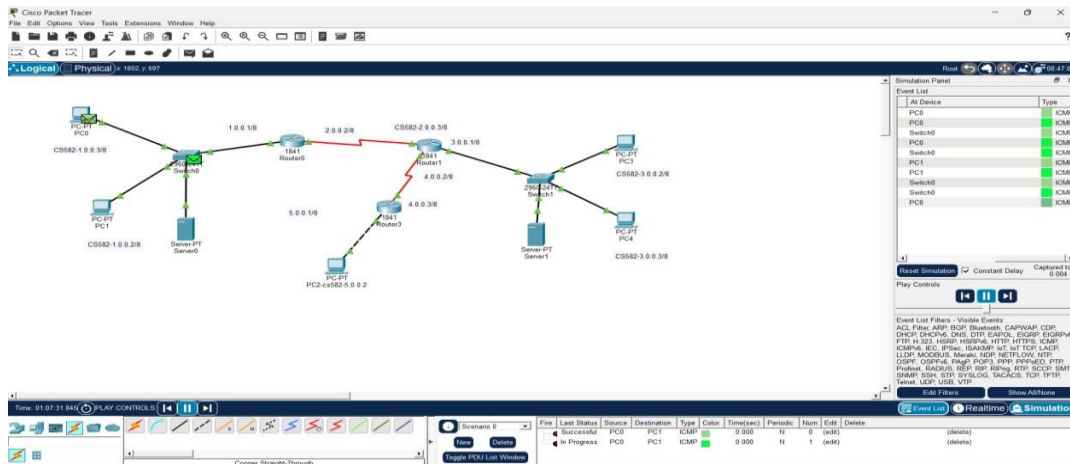
An IP address is required for a newly added interface so that it can identify itself in the network and communicate with other devices. Without an IP address, the interface cannot send or receive data. In this topology, each router interface connects to a different network, so assigning an IP address is necessary for proper routing and communication.

5. What Is a Routing Table and Why Is It Important?

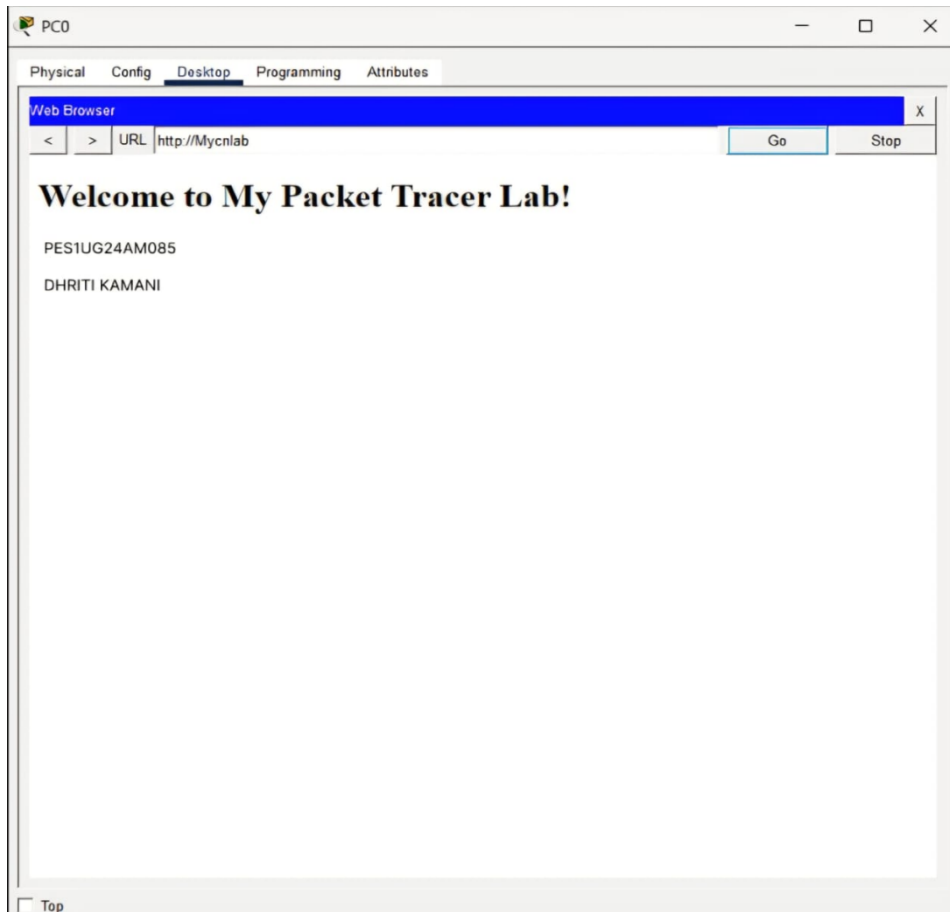
A routing table is a table stored in a router that contains information about different networks and how to reach them. It helps the router decide where to forward data packets. The routing table is important because, without it, the router will not know the correct path to send data to other networks.

TASK 2 (B)

1.



2.



3. What is your observation and understanding of this concept that you have implemented?

From this task, I understood how DNS works in a network. After configuring DNS entries, PCs were able to ping other devices using names instead of IP addresses. The web server was also accessible using the domain name. This showed how DNS makes network communication easier and more user-friendly by translating names into IP addresses.

4. Why Must Each PC Be Configured with the DNS Server's IP Address?

Each PC must be configured with the DNS server IP address so that it knows where to send name resolution requests. Without configuring the DNS server IP, the PC cannot convert domain names into IP addresses, and ping or web access using names will fail.

5. What Is the Role of the DNS Server in a Network?

The DNS server's role is to convert human-readable names into IP addresses. It helps devices locate other systems on the network without remembering IP addresses. DNS is important for accessing services like websites and servers using names instead of numbers.